

Understanding how $Y \mid X$ shifts with PO-risk

a minimal prototype (one model per target metric)

1 Idea

HSIC gives a *symmetric association* between two metrics. But to understand whether the *relationship* between a target metric Y (say GMV) and the other metrics X has *changed* between an existing and a new batch, we need a target-oriented, conditional object, not a pairwise one. We use the pseudo-outcome (PO-)risk with the batch label W as a “treatment”: it detects a change in $P(Y \mid X)$ (concept drift), and a leave-one-out (LOCO) tells us *which* X the change loads on. This is **one model per target** Y — the learner takes all of X jointly — so it costs $O(\#\text{targets})$ models, not $O(p^2)$ models for every X - Y pair.

2 PO-risk

With cross-fitted nuisances $\mu(x) = \mathbb{E}[Y \mid x]$ and $e(x) = \Pr(W=1 \mid x)$, form the pseudo-outcome $\psi = (Y - \mu(X))(W - e(X))$ and fit $\tau = \arg \min_f \mathbb{E}[(\psi - f(X))^2]$. The statistic is

$$\text{PO-risk} = \mathbb{E}[\tau(X)^2], \tag{1}$$

which is large when $\tau(X) = \mathbb{E}[Y \mid X, W=1] - \mathbb{E}[Y \mid X, W=0] \neq 0$, i.e. when $P(Y \mid X)$ differs across batches, and ≈ 0 under pure covariate shift. A permute-then-refit p -value (shuffle W) tests significance.

LOCO. The importance of metric x_j is the drop in PO-risk when it is removed, $\phi_j = \text{PO-risk}(X) - \text{PO-risk}(X_{-j})$: removing the metric that carries the $Y \mid X$ change collapses the detected drift.

3 Method

Choose a target Y ; let X be the other metrics and W the batch. (1) Run the PO-risk permutation test to decide whether $P(Y \mid X)$ shifted. (2) If so, rank the X by LOCO to localize where the change sits. One fit of the meta-learner covers all X jointly.

4 Result

Synthetic metrics are drawn i.i.d. (no covariate shift); the target $Y = \text{GMV}$ keeps a stable mapping but, *for the new batch only*, gains a dependence on **rating** (the $\text{rating} \rightarrow \text{GMV}$ relationship strengthens) — a concept drift localized to **rating**.

<i>PO-risk test</i> (statistic = 0.118, permutation $p = 0.010 \Rightarrow Y \mid X$ shifted)	
metric x_j	LOCO drop ϕ_j
rating (driver)	+0.106
orders	+0.046
delivery	+0.039
promo	+0.032
tenure	+0.028

Table 1: PO-risk with target $Y = \text{GMV}$. The permutation test detects that $P(Y \mid X)$ shifted ($p = 0.010$); the LOCO drop localizes the change to **rating**, whose removal collapses the PO-risk far more than any other metric. One PO-risk model (all X jointly) suffices — no per-pair modelling.

Takeaway. For a target metric, PO-risk answers the question HSIC cannot — *did $Y \mid X$ change, and on which X* — with a single meta-learner per target and a LOCO read-out, avoiding an $O(p^2)$ battery of pairwise models.